

ORF307

Duality I

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Today's topics

- Obtaining lower bounds
- The dual problem
- Weak and strong duality

HW5 is out today and due on Oct. 31, 11:59PM

Please reach out if you want to discuss Midterm 1

- **Recap**

Standard form polyhedra

LP in standard form:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array}$$

The feasible region is a standard polyhedron

$$P = \{x \in R^n \mid Ax = b, x \geq 0\}$$

LP in inequality form:

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

The feasible region is a standard polyhedron

$$P' = \{x \in R^n \mid Ax \leq b\}$$

-
- **Obtaining lower bounds**

Obtaining lower bounds

A simple example:

$$\begin{array}{ll} \min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + 3x_2 \geq 2 \end{array}$$

Question:

What is a **lower bound** on the **optimal cost**?

Observation: the value of the objective cannot be lower than 2 (due to the constraint)

Then: 2 is a lower bound on optimal cost!

Obtaining lower bounds

Another simple example:

$$\begin{array}{ll}\min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & x_2 \geq 1\end{array}$$

Question: What is a **lower bound** on the **optimal cost**?

Observation: if we multiply the 2nd constraint by 2 and add it to the 1st constraint we get the objective function:

$$1 \times (x_1 + x_2 \geq 2) + 2 \times (x_2 \geq 1) \rightarrow x_1 + 3x_2 \geq 4$$

Then: 4 is a lower bound on optimal cost!

Obtaining lower bounds

A more interesting example:

Let's find a lower bound for the LP:

$$\left. \begin{array}{ll} \min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & x_2 \geq 1 \\ & x_1 - x_2 \geq 3 \end{array} \right\} \quad (\mathbf{P})$$

**Not as easy as the
previous examples...**

Obtaining lower bounds

A more interesting example:

Let's find a lower bound for the LP:

$$\begin{array}{ll} \min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & x_2 \geq 1 \\ & x_1 - x_2 \geq 3 \end{array} \quad \left. \vphantom{\begin{array}{l} \min \\ \text{s.t.} \end{array}} \right\} \quad (\mathbf{P})$$

Multiply/Add constraints (multipliers y_i must be nonnegative):

$$y_1 \cdot (x_1 + x_2 \geq 2) + y_2 \cdot (x_2 \geq 1) + y_3 \cdot (x_1 - x_2 \geq 3) \Rightarrow \underbrace{(y_1 + y_3)x_1 + (y_1 + y_2 - y_3)x_2}_{\text{Cost}} \geq \underbrace{2y_1 + y_2 + 3y_3}_{\text{Lower Bound}}$$

Match cost coefficients:

$$y_1 + y_3 = 1, \quad y_1 + y_2 - y_3 = 3, \quad y_1, y_2, y_3 \geq 0$$

Many options for the multipliers $\rightarrow$ **many lower bounds:**

$$y = (1, 2, 0)^T \Rightarrow 2y_1 + y_2 + 3y_3 = 4$$

$$\bar{y} = (0, 4, 1)^T \Rightarrow 2y_1 + y_2 + 3y_3 = 7$$

How can we find the best multipliers? ₉

Obtaining lower bounds

A more interesting example:

How can we find the
best multipliers?

The optimal solution of the LP (maximization) gives us the optimal multipliers y

$$\left. \begin{array}{ll} \text{max} & 2y_1 + y_2 + 3y_3 \\ \text{s.t.} & y_1 + y_3 = 1 \\ & y_1 + y_2 - y_3 = 3 \\ & y_1, y_2, y_3 \geq 0 \end{array} \right\} \quad \text{(D)}$$

$$\left. \begin{array}{ll} \text{min} & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & x_2 \geq 1 \\ & x_1 - x_2 \geq 3 \end{array} \right\} \quad \text{(P)}$$

Remark: problem (D) is called the dual problem of problem (P)

Remark: take a moment and observe the objective and constraints of these problems.

Do you see any relation between the coefficients of the constraints?

Hint: Check the constraint coefficient matrices, the rhs, and the cost coefficients

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- **The dual problem**

Primal max and Dual min (different perspectives – both optimize)

- **The primal problem:** Maximize the profit of your bike company
 - You produce 3 types of bikes and when you **sell** unit of each type you earn a specific profit
 - You use 3 types of **resources with different costs** to produce the bikes (e.g., aluminum, plastic, labor)
 - You want to **maximize your profit**
- **The Dual problem:** Minimize the value (or cost) of the resources
 - I want to **buy** all your resources and produce the bikes for you (or for me)
 - I need to ensure the **total value** of the needed for resources used to make one unit of every bike type is at least the profit you get from selling that bike
 - I want to find the **minimum total value** (or cost) for all types of resources and then make my offer.

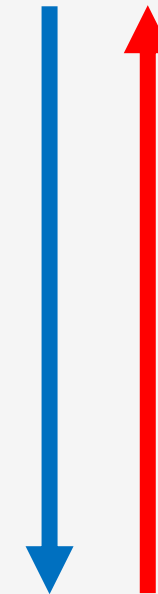
Primal and Dual LPs (different perspectives- same problem)

- **The primal problem:** Maximize your profit

$$\begin{array}{ll}\max & 400x_1 + 600x_2 + 800x_3 & \text{(Total profit)} \\ \text{s.t.} & 5x_1 + 8x_2 + 12x_3 \leq 200 & \text{(Aluminum constraint)} \\ & 10x_1 + 8x_2 + 15x_3 \leq 100 & \text{(Labor constraint)} \\ & 4x_1 + 5x_2 + 6x_3 \leq 80 & \text{(Plastic constraint)} \\ & x_1, x_2, x_3 \geq 0\end{array}$$

- **The Dual problem:** Minimize the value (or cost) of the resources

$$\begin{array}{ll}\min & 200y_1 + 100y_2 + 80y_3 & \text{(Total resource value)} \\ \text{s.t.} & 5y_1 + 10y_2 + 4y_3 \geq 400 & \text{(Bike type A constraint)} \\ & 8y_1 + 8y_2 + 5y_3 \geq 600 & \text{(Bike type B constraint)} \\ & 12y_1 + 15y_2 + 6y_3 \geq 0 & \text{(Bike type C constraint)} \\ & y_1, y_2, y_3 \geq 0\end{array}$$

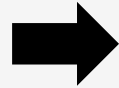


***We will learn how to
Derive the dual problem
Of every LP problem***

Lagrange multipliers

LP in standard form:

$$\left. \begin{array}{l} \min c^T x \\ \text{s.t. } Ax = b \\ x \geq 0 \end{array} \right\} \quad (1)$$



Relax the equality constraints:

$$\left. \begin{array}{l} g(y) = \min_x c^T x + y^T (Ax - b) \\ \text{s.t. } x \geq 0 \end{array} \right\} \quad (2)$$

$g(y)$ is the **optimal objective** of (2)
for **some fixed** multiplier vector y

Assume x^* is the **optimal solution** of (1).

Then we must have: $Ax^* = b$, $x^* \geq 0$.

Finding the best lower bound

Since $g(y)$ is the optimal objective value of problem (2) for some y , we must have:

$$g(y) = \min_x c^T x + y^T (Ax - b) \leq c^T x^* + y^T \underbrace{(Ax^* - b)}_{= 0} = c^T x^*, \quad \forall y$$

Best lower bound is found by solving the maximization problem:

$$\max_y g(y)$$

(we are looking the **largest possible lower bound** for any value of the vector y)

The dual problem

The dual function:

$$\begin{aligned} g(y) &= \min_{x \geq 0} (c^T x + y^T (Ax - b)) \\ &= -y^T b + \min_{x \geq 0} (c^T x + y^T Ax) \\ &= -y^T b + \min_{x \geq 0} (c + A^T y)^T x \end{aligned}$$

Does not depend on x

We can identify **two cases** for $g(y)$:

$$g(y) = \begin{cases} -y^T b, & \text{if } c + A^T y \geq 0 \\ -\infty, & \text{if } c + A^T y < 0 \end{cases}$$

The dual problem:

$$\begin{aligned} \max_y g(y) &= \max_y -b^T y \\ \text{s.t. } & A^T y + c \geq 0 \end{aligned}$$

Summary: Primal and Dual problems

Primal problem

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

$x \in R^n$ are called the **primal variables**

Dual problem

$$\begin{aligned} \max_{y} \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c \geq 0 \end{aligned}$$

$y \in R^m$ are called the **dual variables**

The dual problem carries **very useful information** for the primal problem
(recall the bike problem presented in slide 13)

Duality is also useful to solve optimization problems in general

Dual of LP with inequality constraints

Primal problem in inequality form

$$\left. \begin{array}{l} \min c^T x \\ \text{s.t. } Ax \leq b \end{array} \right\} \quad (1)$$

Assume x^* is the **optimal solution** of (1).

Then we must have: $Ax^* \leq b$

Relax the equality constraints:

$$g(y) = \min c^T x + y^T (Ax - b) \quad (2)$$

$g(y)$ is the optimal objective of (2)
for some multiplier vector y

Finding the best lower bound

Since $g(y)$ is the optimal objective value of problem (2) for some $y \geq 0$, we must have:

$$g(y) = \min c^T x + y^T (Ax - b) \leq c^T x^* + \underbrace{y^T (Ax^* - b)}_{\leq 0} \leq c^T x^*, \quad \forall y \geq 0$$

*The only differences
from slide 14*

Best lower bound is found by solving the maximization problem:

$$\max_{y \geq 0} g(y)$$

(we are looking the **largest possible lower bound** for any value of the vector $y \geq 0$)

Dual of LP *with inequality constraints* – Derivations

The dual function:

$$g(y) = \min(c^T x + y^T (Ax - b))$$

$$= -y^T b + \min(c^T x + y^T Ax)$$

Does not depend on x

$$= -y^T b + \min (c + A^T y)^T x$$

Hence the dual function has the form:

$$g(y) = \begin{cases} -y^T b, & \text{if } c + A^T y = 0 \text{ and } y \geq 0 \\ -\infty, & \text{if otherwise} \end{cases}$$

The dual problem:

$$\begin{aligned} \max_y g(y) &= \max_y -b^T y \\ \text{s.t. } & A^T y + c = 0 \\ & y \geq 0 \end{aligned}$$

Summary: Primal and Dual problems

Primal problem

$$\begin{aligned} \min \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

$x \in R^n$ are called the **primal variables**

Dual problem

$$\begin{aligned} \max_{y} \quad & -b^T y \\ \text{s.t.} \quad & A^T y + c = 0 \\ & y \geq 0 \end{aligned}$$

$y \in R^m$ are called the **dual variables**

General forms

Standard form LP

Primal

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax = b \\ & x \geq 0\end{array}$$

Dual

$$\begin{array}{ll}\max_{y} & -b^T y \\ \text{s.t.} & A^T y + c \geq 0\end{array}$$

Inequality form LP

Primal

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

Dual

$$\begin{array}{ll}\max_{y} & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$

LP with Equalities and Inequalities

Primal

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b \\ & Cx = d\end{array}$$

Dual

$$\begin{array}{ll}\max_{y,z} & -b^T y - d^T z \\ \text{s.t.} & A^T y + C^T z + c = 0 \\ & y \geq 0\end{array}$$

Derive it as exercise! 20

Previously studied example (slide 10) - revisited

Primal

$$\begin{array}{ll}\min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2 \\ & x_2 \geq 1 \\ & x_1 - x_2 \geq 3\end{array}$$

Inequality form LP

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

Convert $\geq$ to $\leq$ and then formulate the Dual

Data

$$c = (1, 3)^T, \quad b = (-2, -1, -3)^T$$

$$A = \begin{bmatrix} -1 & -1 \\ 0 & -1 \\ -1 & 1 \end{bmatrix}$$

Dual

$$\begin{array}{ll}\max_y & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$



$$\begin{array}{ll}\max_y & 2y_1 + y_2 + 3y_3 \\ \text{s.t.} & -y_1 - y_3 = -1 \\ & -y_1 - y_2 + y_3 = -3 \\ & y_1, y_2, y_3 \geq 0\end{array}$$

To memorize

Ways to easily get the dual problem:

- Derive dual function directly
- Transform the problem in inequality form LP and dualize

Sanity checks and signs conversion

- Consider constraints as $Ax - b \leq 0$ or $Ax - b = 0$ (**never as** ≥ 0 *minimize things to remember!*)
- y **free** ($\in R^m$) for primal **equalities** and $y \geq 0$ for primal **inequalities**

Dual of the Dual

Theorem:

Assume you have an LP (primal) and compute its dual LP. Then if you take the dual of the dual LP you will obtain the original (primal) LP.

Exercise

Derive the dual and then dualize it

Primal	Dual	Dual of the dual
$\min c^T x$ s.t. $Ax \leq b$ $Cx = d$	$\max -b^T y - d^T z$ s.t. $A^T y + C^T z + c = 0$ $y \geq 0$	?

Theorem:

If we transform an LP (e.g., inequality form) to another equivalent form (e.g., standard form) and then form their duals, then the dual problems of both of these problems are equivalent.

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- **Weak and strong duality**

Optimal objective values

Primal

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax \leq b\end{array}$$

p^* is the primal optimal value

If primal is infeasible $\rightarrow p^* = +\infty$

If primal is unbounded $\rightarrow p^* = -\infty$

Dual

$$\begin{array}{ll}\max_{y} & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0\end{array}$$

d^* is the primal optimal value

If dual is infeasible $\rightarrow d^* = -\infty$

If dual is unbounded $\rightarrow d^* = +\infty$

Weak duality

Theorem: if $x \in R^n$ and $y \in R^m$ satisfy

- x is a feasible solution to the **Primal**
 - y is a feasible solution to the **Dual**
- $$\left. \vphantom{\begin{matrix} \bullet \\ \bullet \end{matrix}} \right\} \Rightarrow -b^T y \leq c^T x \quad (\text{when we have minimization Primal problem})$$

Proof:

Since x is feasible for the Primal: $Ax \leq b \Rightarrow y^T Ax \leq y^T b$.
Since y is feasible for the Dual: $Ay + c = 0$ and $y \geq 0$.

$$\left. \vphantom{\begin{matrix} \text{Since } x \text{ is feasible} \\ \text{Since } y \text{ is feasible} \end{matrix}} \right\} \Rightarrow y^T Ax \leq y^T b \Rightarrow -c^T x \leq b^T y \Rightarrow -b^T y \leq c^T x$$


Remarks:

- For any dual feasible solution y , the dual objective $-b^T y$ is a **lower bound** to the **primal objective** $c^T x$
- For any primal feasible solution x , the primal objective $c^T x$ is an **upper bound** to the **dual objective** $-b^T y$

Duality gap: $c^T x - (-b^T y) = c^T x + b^T y \rightarrow$ the **difference** of the **primal** and **dual** objectives

Week duality – Corollaries

Unboundedness –vs- feasibility

- Primal unboundedness ($p^* = -\infty$) $\Rightarrow$ Dual infeasibility ($d^* = -\infty$)
- Dual unboundedness ($d^* = +\infty$) $\Rightarrow$ Primal infeasibility ($p^* = +\infty$)

Optimality conditions

IF x, y satisfy:

- x is feasible solution to the primal LP
- y is feasible solution to the dual LP
- The duality gap is zero (i.e., $g = c^T x + b^T y = 0$)

THEN x is an optimal solution to the Primal and y is an optimal solution to the Dual.

Strong duality

Theorem:

If an LP has an optimal solution x^* then its dual LP **has also** an optimal solution y^* , and the optimal objective values of the Primal and Dual LPs are **equal**, i.e.,

$$p^* = c^T x^* = -b^T y^* = d^*$$

In such a case the **duality gap is zero** ($c^T x^* + b^T y^* = 0$).

Proof: next slide →

Strong duality

Proof:

Assume that x^* is optimal solution to the Primal. We will **construct** an optimal solution y^* for the dual.

Apply the Simplex method to the LP (in standard form):

$$\left. \begin{array}{l} \min c^T x \\ \text{s.t. } Ax = b \\ x \geq 0 \end{array} \right\} \longrightarrow \begin{array}{l} \bullet \text{ Optimal basis: } B \\ \bullet \text{ Optimal solution: } x^* \text{ with } A_B x_B^* = b \\ \bullet \text{ Reduced costs: } \bar{c} = c - A^T A_B^{-T} c_B \geq 0 \Rightarrow -A^T A_B^{-T} c_B \geq -c \end{array}$$

Define y^* such that $y^* = -A_B^{-T} c_B$. Then $A^T y^* = -A^T A_B^{-T} c_B \geq -c \Rightarrow A^T y^* + c \geq 0$ (y^* **dual feasible**)

$$-b^T y^* = -b^T (-A_B^{-T} c_B) = -(A_B x_B^*)^T (-A_B^{-T} c_B) = x_B^{*T} A_B^T A_B^{-T} c_B = c_B^T x_B^* = c^T x^*$$

Hence the **duality gap**: $c^T x^* + b^T y^* = 0$.

By the weak duality corollary, we have that y^* is dual optimal.



Exception to strong duality

Primal

$$\begin{array}{ll}\min & x_1 + 2x_2 \\ \text{s.t.} & x_1 + x_2 = 1 \\ & 2x_1 + 2x_2 = 3\end{array}$$

Infeasible LP.

Optimal value: $p^* = +\infty$

Dual

$$\begin{array}{ll}\max_y & -y_1 - 3y_2 \\ \text{s.t.} & y_1 + 2y_2 = 1 \\ & y_1 + 2y_2 = 2\end{array}$$

Infeasible LP.

Optimal value: $d^* = -\infty$

Both Primal and Dual problems are infeasible

Relationships between Primal and Dual problems

Primal \ Dual	$p^* = +\infty$ (infeasible)	p^* finite (optimal)	$p^* = -\infty$ (unbounded)
$d^* = +\infty$ (unbounded)	primal inf. dual unb.	Impossible	Impossible
d^* finite (optimal)	Impossible	optimal values equal	Impossible
$d^* = -\infty$ (infeasible)	exception	Impossible	primal unb. dual inf

- Upper-right excluded by weak duality
- (1,1) and (3,3) are proven using weak duality
- (3,1) and (2,2) are proven using strong duality

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- **Example**

Production optimization

$$\begin{array}{ll} \text{max} & x_1 + 2x_2 \quad \leftarrow \text{Profit by selling products} \\ \text{s.t.} & \left. \begin{array}{l} x_1 \leq 100 \\ 2x_2 \leq 200 \\ x_1 + x_2 \leq 150 \\ x_1, x_2 \geq 0 \end{array} \right\} \leftarrow \text{Resource constraints} \end{array}$$

Dualize:

1. **Transform** the **max** into a **min** LP:

$$\begin{array}{ll} \text{min} & c^T x \\ \text{s.t.} & Ax \leq b \end{array} \quad \begin{array}{l} \text{red } \Updownarrow \\ c = \begin{pmatrix} -1 \\ -2 \end{pmatrix}, \quad A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad b = \begin{pmatrix} 100 \\ 200 \\ 150 \\ 0 \\ 0 \end{pmatrix} \end{array}$$

2. Derive the **dual** of the **min** LP:

$$\begin{array}{ll} \text{max}_y & -b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0 \end{array} \quad \Leftrightarrow \quad \begin{array}{ll} \text{min}_y & b^T y \\ \text{s.t.} & A^T y + c = 0 \\ & y \geq 0 \end{array}$$

Production optimization

Dual LP: $\max_y -b^T y$
s.t. $A^T y + c = 0$
 $y \geq 0$

$\Leftrightarrow$

$\min_y b^T y$
s.t. $A^T y + c = 0$
 $y \geq 0$

$c = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$, $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$, $b = \begin{pmatrix} 100 \\ 200 \\ 150 \\ 0 \\ 0 \end{pmatrix}$

Final dual

$$\min 100y_1 + 200y_2 + 150y_3$$

$$\begin{aligned} \text{s.t. } & y_1 + y_3 - y_4 = 1 \\ & 2y_2 + y_3 - y_5 = 2 \\ & y_1, y_2, y_3, y_4, y_5 \geq 0 \end{aligned}$$



y_4, y_5 can be thought of as slack variables.
They can be eliminated and replaced by $\geq$

Final dual LP (fewer vars)

$$\min 100y_1 + 200y_2 + 150y_3$$

$$\begin{aligned} \text{s.t. } & y_1 + y_3 \geq 1 \\ & 2y_2 + y_3 \geq 2 \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Production optimization

$$\begin{aligned} \textbf{Dual LP:} \quad & \min \quad 100y_1 + 200y_2 + 150y_3 \\ & \text{s.t.} \quad y_1 + y_3 \geq 1 \\ & \quad \quad 2y_2 + y_3 \geq 2 \\ & \quad \quad y_1, y_2, y_3 \geq 0 \end{aligned}$$

Interpretation

- Sell all your resources at a fair (minimum) price
- Selling must be more convenient than producing:
 - **Product 1:** (price 1, needs 1 × resource 1 and 3): $y_1 + y_3 \geq 1$
 - **Product 2:** (price 2, needs 2 × resource 1 and 1 × resource 3): $2y_2 + y_3 \geq 2$

References

- Bertsimas/Tsitsiklis: Introduction to Linear Optimization
 - Ch. 4: Duality theory
- Vandebei: Linear Programming: Foundations and Extensions
 - Ch. 5: Duality theory

Next lecture

- More on duality:
 - Game theory
 - Complementarity slackness
 - Farkas lemma

